Jennifer Ren

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EDUCATION

B.S. in Electrical Engineering and Computer Sciences — *Aug 2024 – May 2028*, *Berkeley, CA*

UNIVERSITY OF CALIFORNIA, BERKELEY | GPA: 3.79/4.00

Coursework: Data Structures & Algorithms, Computer Architecture, Digital Design & IC, Machine Learning, Internet Protocols, Computer Security, Signals & Systems, Circuits & Devices, Efficient Algorithms & Intractable Problems, Discrete Math & Probability, Linear Algebra & Differential Equations, Multivariable Calculus

Interests: Graphic Design/Painting, Competitive Figure Skating, Poker

SKILLS

Languages	SystemVerilog, Verilog, Python, C, C++, Java, x86/RISC-V Assembly, LaTeX
EDA Tools	Synopsys VCS, Synopsys Verdi, Cadence Genus, Cadence Innovus, Cadence Joules
Other Tools	LTspice, Logisim, MATLAB, Oscilloscope, Spectrum Analyzer

PROJECTS

3-Stage Pipelined RISC-V CPU (EECS151) — *2026*

SYSTEMVERILOG | SYNOPSYS VCS, VERDI | CADENCE GENUS, INNOVUS, JOULES

- Designed a 3-stage pipelined RISC-V CPU in SystemVerilog
- Implemented hazard detection, forwarding, stalls, and branch/jump flushing
- Verified with assembly tests and post-synthesis simulation; ran synthesis, place-and-route, and power analysis
- Placed 1st in the final project's leader board by frequency, with a top 10 figure of merit

CO_2 Database Query System with Visualization — *2024*

PYTHON | SQL | MATLAB | BEAUTIFUL SOUP

- Scraped climate data, stored it in a structured SQL database, and built a server-client query system
- Visualized trends in MATLAB with scatter plots, histograms, and linear regression

EXPERIENCE

Undergraduate Researcher, AI-Driven Tapeout — *Jun 2026 – Present*, *Berkeley, CA*

UC BERKELEY EECS | DIGITAL DESIGN, ML FOR EDA

- Investigating applications of machine learning to accelerate RTL-to-GDSII tapeout flows under Professor Sagar Karandikar

Course Staff, EECS 151 (Digital Design & Integrated Circuits) — *Aug 2026 – Dec 2026*, *Berkeley, CA*

UC BERKELEY EECS DEPARTMENT

- Selected as UCS1 tutor for Berkeley's core digital design course covering SystemVerilog, CMOS, RTL design, and full ASIC/FPGA flow
- Will lead lab sections and office hours supporting students through ASIC-based RISC-V CPU design and synthesis labs

Machine Learning Intern — *Jan 2026 – Present*

BFAI SEMICONDUCTOR | PYTHON, PYTORCH, COMPUTER VISION

- Developed ML pipelines for semiconductor defect detection using classification, detection, and segmentation models
- Built preprocessing and augmentation workflows for crack and void detection in Docker containers

Project Lead, Cal Women's Network Mentorship Program — *Aug 2025 – Dec 2025*, *Berkeley, CA*

BERKELEY DATA DISCOVERY

- Cleaned and analyzed membership data using SQL, pandas, and Matplotlib
- Prototyped a mentorship platform and matching algorithm for 200+ members

Senior Mentor, CSM16A — *Jan 2025 – Present*, *Berkeley, CA*

COMPUTER SCIENCE MENTORS AT BERKELEY

- Led tutoring sessions and created supplemental notes and resources for 100+ students
- Supported students in signals, linear algebra, and data processing concepts